

I Part-Segmented 3D Asset Collection

Data Sources

- PartVerse
- PartObjaverse-Tiny
- Public 3D sources (Objaverse, ...)



Part-segmented Mesh (≤ 16 parts)

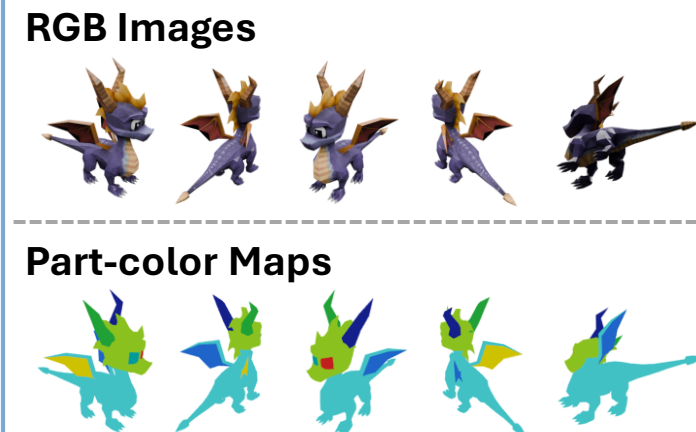


II Part-Label Refinement & Multi-view Rendering

II-a. Part Semantic Labeling

Part ID	Semantic Label	RGB
part_01	horn (right)	[30, 160, 50]
part_02	horn (left)	[20, 30, 130]
part_03	head	[130, 220, 30]
...	...	...

II-b. Multi-view Rendering



III Semantic Edit Planning

III-a. Part Grouping

- horn (right)
[30, 160, 50]
- horn (left)
[20, 30, 130]

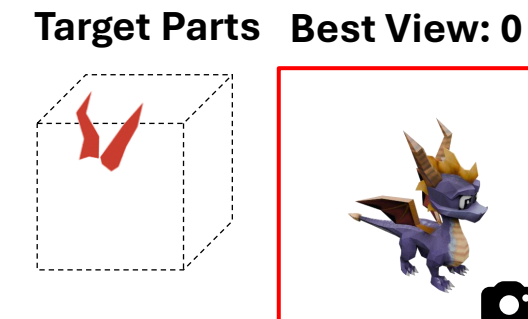
III-b. Edit Instruction Generation



Qwen-3.5

Instruction:
Replace the textured horns with curved ivory tusks.
Edit Type: Replace

III-c. Target Parts & View Selection



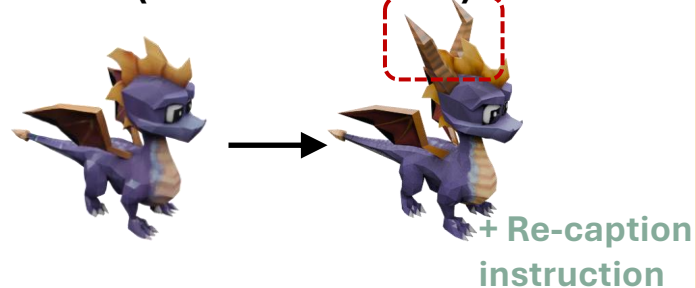
IV Type-aware Edit Planning & Verification

IV-a. Deletion / Addition (3D)

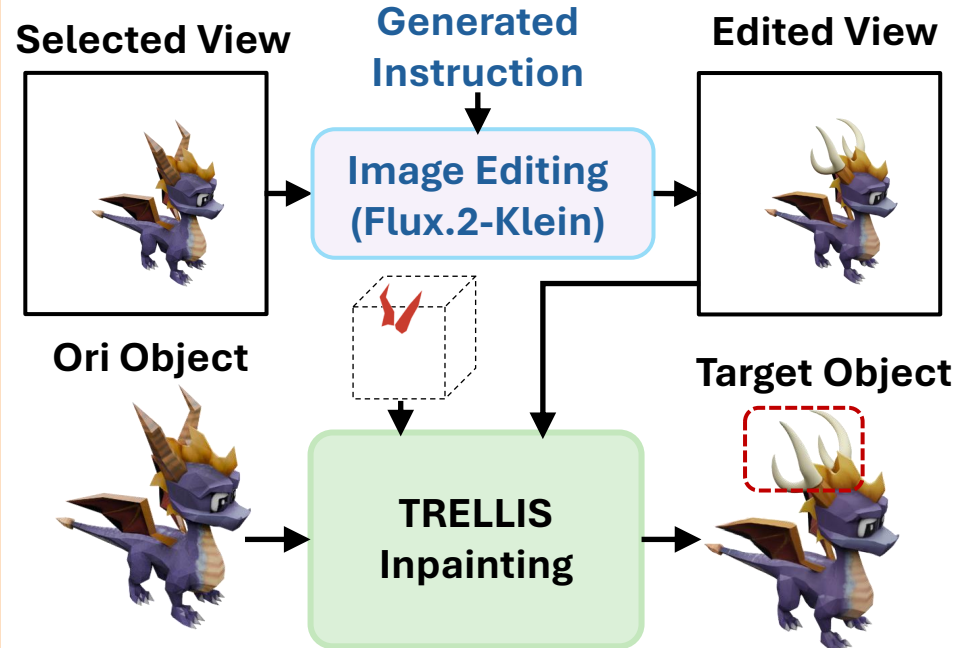
Deletion (remove the parts in 3D)



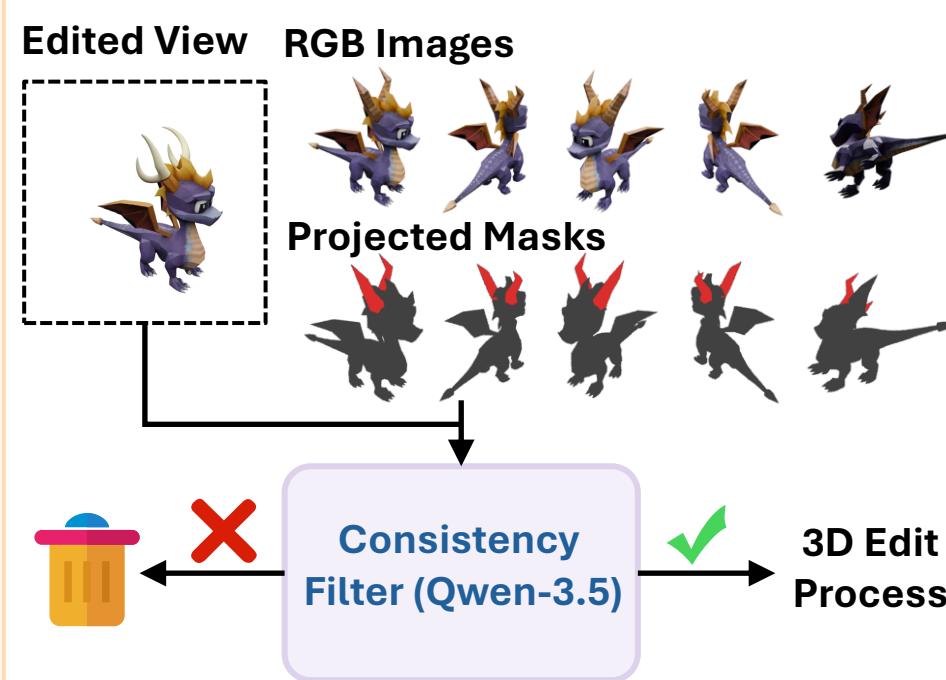
Addition (reverse deletion)



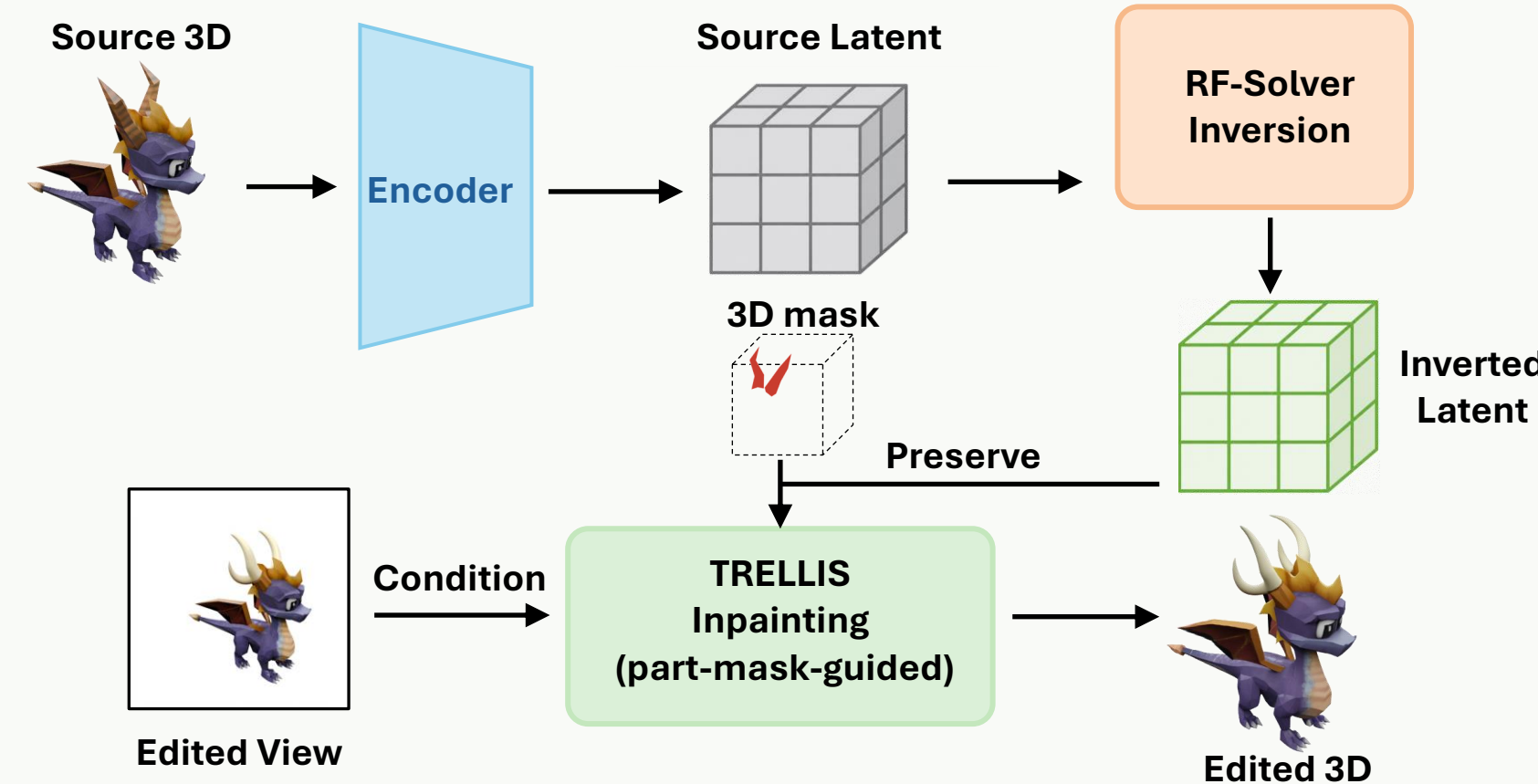
IV-b. Replace / Scale / Material / Color / Style (2D→3D)



IV-c. Mid-verification



V Region-controlled 3D Transfer



VI Multi-view Quality Verification

Inputs

- Before (5 views)
- After (5 views)
- Edit instruction
- Target part
- Target description



Automatic Checks (Qwen-3.5)

- Edit executed as instructed
- Correct target region edited
- Unedited regions preserved
- Visual quality is good
- Artifact-free and consistent

Decision

- Pass
- Keep
- Fail
- Keep
- Discard / Regenerate

VII Pxform Editing Pair Output

Dataset Statistics

- 102,007 Training Pairs
- 1,497 Test Pairs
- 11,273 Source Objects
- 7 Edit Types

